A New Smallest 1200V Intelligent Power Module for Three Phase Motor Drives

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***Abstract*— This paper presents the new smallest 1200V Intelligent Power Module (IPM) of 5A and 10A rating an excellent solution for three phase AC motors and permanent magnet motors in variable speed drives applications such as fan drives, active filter for HVAC and low power motor drives of GPI and Servo drives. This IPM integrates six IGBTs and six diodes with 6 channel Silicon On Insulator (SOI) gate driver in Dual in line (DIP) package with Direct Bond Copper (DBC). This paper provides an overall description regarding electrical characteristics, device performance and package.**

***Keywords*— Infineon, CIPOS™, IPM, Three phase Motor drives, Inverter**

I. Introduction

The importance of energy saving with a regard to environmental issues has been grown bigger and the inverters are increasingly used for applications of a wide range. The new 1200V IPM has been designed and developed in a compact size package of Infineon Technologies with newly developed SOI single gate driver IC. This IPM is called Control Integrated Power System (CIPOS™) Maxi and it is optimized to improve efficiency, reliability and controllability within industrial motor drive applications. This paper describes the features of the internal components as well as the package structure and thermal performance.

II. Features of Design and Electrical Performance

A. Overview and circuit configuration

Figures 1 and 2 show the package drawing and the internal block diagram of the CIPOS™ Maxi respectively. It has been designed for the smallest package size as 1200V IPM without dummy pin by optimized internal PCB for gate driver IC and optimized DBC for power devices. CIPOS™ Maxi is composed of 6 IGBTs, 6 diodes in a three phase inverter structure together with a new 1200V SOI single gate driver IC. Especially this single gate driver IC with integrated bootstrap circuit and several protection functions paves the way for minimization. The package development targets the smallest possible size which still meets international industrial standards for insulation distance such as clearance and creepage. CIPOS™ Maxi devices are available with rated current of 5A and 10A, and each device names are listed in the Table I.

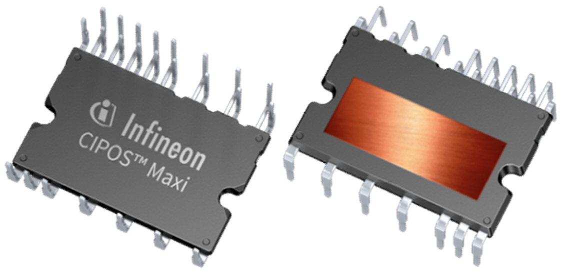

Fig. 1. Package drawing (size: 36mm x 22.7mm x 3.1mm)

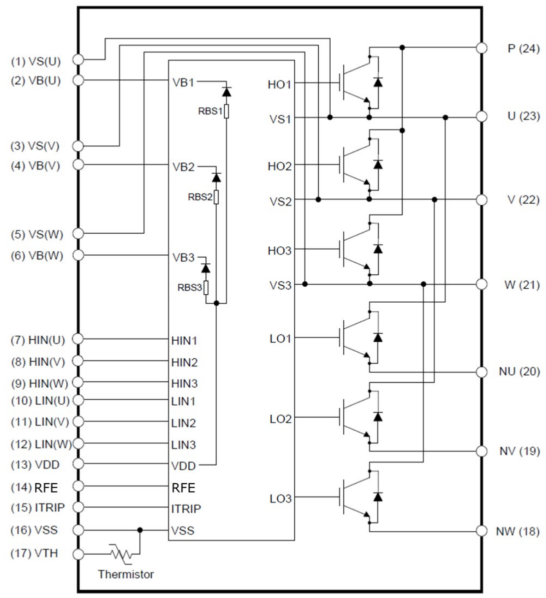

Fig. 2. Internal block diagram

TABLE I

Current rating	Device name	Package
5A	IM818-SCC	24pin, Dual in line
10A	IM818-MCC	

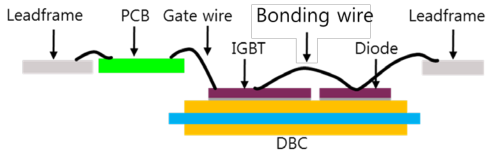

Fig. 3. Internal structure of CIPOS™ Maxi

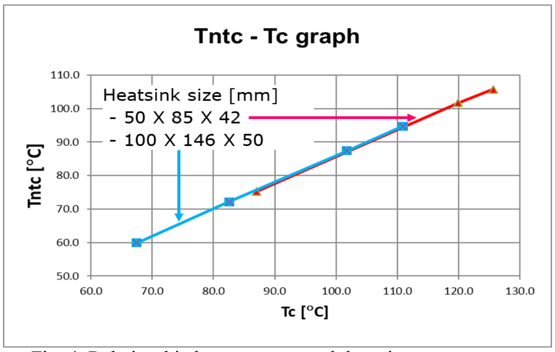

Fig. 4. Relationship between case and thermistor temperature

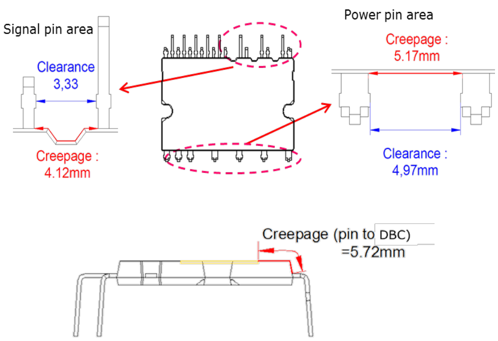

Fig. 5. Clearance and Creepage distance

B. Package

The new package for CIPOS™ Maxi is designed with smallest package size (36mm x 22.7mm x 3.1mm) without dummy pin by optimized internal PCB and DBC structure as shown in Figure 3. Gate driver IC and thermistor are placed on an internal PCB. The IGBTs and diodes are placed on a DBC for higher thermal performance. It adopts Al wire bonding technology for electrical connections between PCB and DBC as well as DBC and Lead-frame. The CIPOS™ Maxi has the independent V_{TH} pin and it is connected to thermistor inside the package, which offers the temperature monitoring function. The thermistor is a NTC type and the case temperature can be tracked by NTC temperature as shown in Figure 4. It is noted that this temperature relationship between NTC and case should be measured on the experimental environment because its relationship can be different according to user's heat dissipation conditions. Also, it is designed with transfer molding technology encapsulation of the internal body and meets all international industrial standards such as clearance and creepage for insulation distance. Figure 5 depicts the insulation distance of pin to pin and pin to DBC.

C. Features of new 1200V SOI single gate driver IC

The new 1200V SOI single gate driver with integrated bootstrap circuit is used to achieve higher levels of integration, reliability and performance in the CIPOS™ Maxi. This SOI gate driver IC disables leakage or latch-up current between adjacent devices structurally. It prevents the latch-up effect even in case of high dV/dt switching and surge under elevated temperature [1]. This gate driver provides several protection functions such as cross-conduction prevention, under voltage lock out, over current detection and enable input. Figure 6 shows the characteristics of the integrated bootstrap circuit.

CIPOS™ Maxi provides an integrated fault output with sleep function and adjustable fault clear time by the RFE pin. There are two situations that would cause the driver IC to report a fault via the RFE pin. The first is an under voltage condition of V_{DD} and the second is if the ITRIP function recognizes a fault. Once the fault condition occurs, the RFE pin is internally pulled to V_{SS} and the fault clear timer is activated; all outputs of this driver are shut down. The RFE output stays in the low state until the fault condition has been removed and the built-in fault clear timer expires which is set to a minimum of 100µs. Once the built-in fault clear timer expires, the voltage on the RFE pin will return to its external pull-up voltage. The output will remain disabled and the fault condition maintained until the voltage on the RFE pin charges up to enable threshold voltage. The charging characteristics is dictated by RC time constant attached to the RFE pin. Figure 7 shows that R_{ext} is connected between the external supply and the RFE pin, while C_{ext} is placed between the RFE and VSS pins. The fault clear time is determined by the charging characteristics of the capacitor where the time constant is set by R_{ext} and C_{ext}.

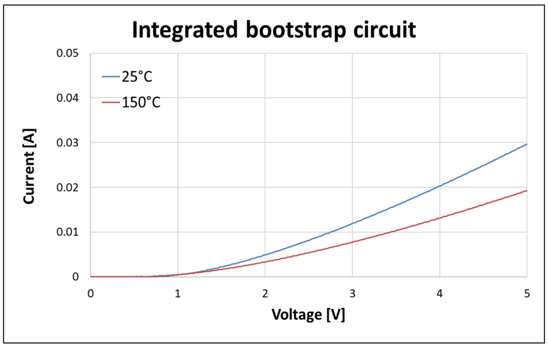

Fig. 6. Bootstrap circuit characteristics

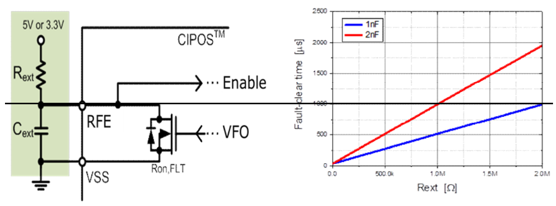

Fig. 7. Circuit for fault-clear time and Adjustable fault-clear time

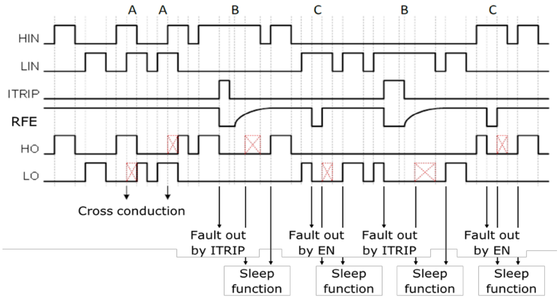

Fig. 8. Timing chart

Figure 8 illustrates that how the gate driver IC operates to protect device from several abnormal situations by the timing chart regarding protection function such as cross-conduction prevention and sleep function. During interval A, the gate driver received the command to turn-on both the high side and low side switches at the same time, the shoot-through protection has prevented this condition. It is keeping on output channel that is already on ignoring the 2nd input signal. Interval B and C shows the sleep function to protect destruction of IGBTs from unexpected and incorrect operation of the micro controller. [2] After the fault clear time, the driver IC is waiting for a new input signal on LIN/HIN before activating the output stage (LO/HO).

D. Performance

CIPOS™ Maxi offers enhanced operating conditions compared to existing 1200V IPMs such as shortest dead time and shortest input pulse width. Recommend minimum dead time is 0.5μs and minimum pulse width is 1μs.

Short circuit capability is guaranteed up to 10μs under condition of V_{DC}=800V, V_{DD}=15V and T_J≤150°C. Figure 9 is measured test waveform of an IM818-MCC under condition of V_{DC}=800V, V_{DD}=15V and T_C=150°C. Figure 10 records actual test results regarding short circuit withstand time without any damage under worse conditions such as higher V_{DC} and V_{DD}.

Figure 11 shows allowable maximum operating current by case temperature (T_C) and Figure 12 shows allowable maximum operating current as a function of PWM carrier frequency (f_{SW}) under condition of V_{DC}=600V, V_{DD}=15V, Power Factor (PF)=0.8, Modulation Index(MI)=0.8, Output frequency(FO)=60Hz and SVPWM.

These are simulated by CIPOSIM, a simulation software to estimate losses, junction temperature and allowable maximum operating conditions, provided by Infineon Technologies.

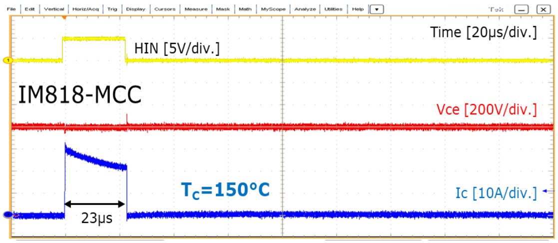

Fig. 9. Short circuit test result of IM818-MCC

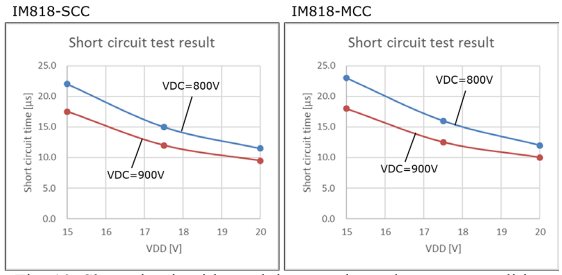

Fig. 10. Short circuit withstand time results under worse condition

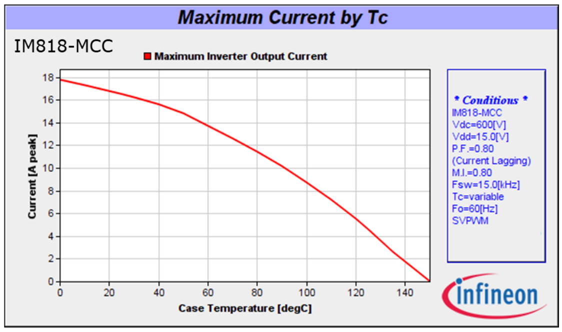

Fig. 11. Allowable maximum current by T_C (V_{DC}=600V, V_{DD}=15V, PF=0.8, MI=0.8, f_{SW}=15 kHz, FO=60Hz, SVPWM)

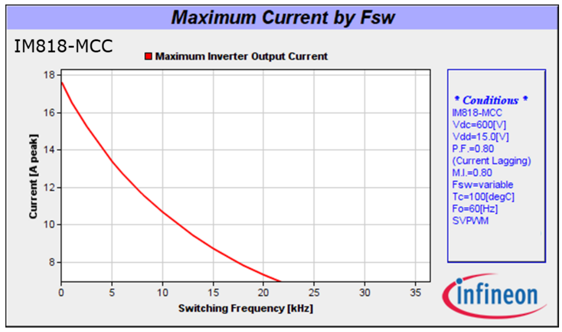

Fig. 12. Allowable maximum current by f_{SW} (V_{DC}=600V, V_{DD}=15V, PF=0.8, MI=0.8, T_C=80°C, FO=60Hz, SVPWM)

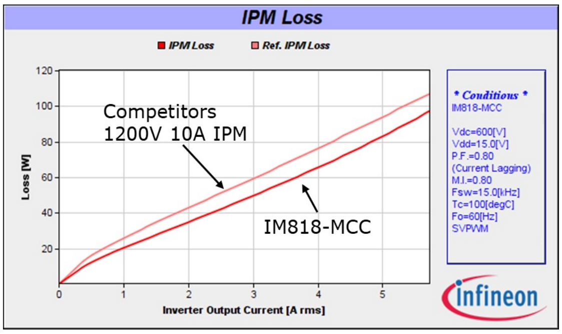

Fig. 13. Inverter loss (V_{DC}=600V, V_{DD}=15V, PF=0.8, MI=0.8, T_C=100°C, f_{SW}=15kHz, FO=60Hz, SVPWM)

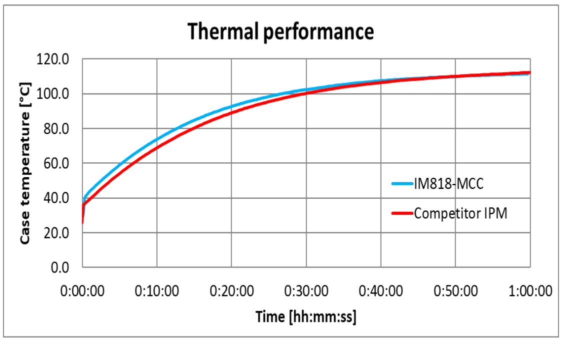

Fig. 14. Thermal performance results (V_{DC}=600V, V_{DD}=15V, PF=1, MI=0.46, Ta=25°C, f_{SW}=15kHz, FO=60Hz, SVPWM)

Figure 13 shows the inverter losses of IM818-MCC compared with competitors 1200V 10A rating IPM under condition of V_{DC}=600V, V_{DD}=15V, PF=0.8, MI=0.8, T_C=100°C, f_{SW}=15kHz, FO=60Hz and SVPWM. IM818-MCC has slightly higher performance at 5Apeak. Figure 14 records evaluation results at using the same heatsink and IM818-MCC shows almost similar thermal performance at the same operating conditions in spite of the package has about half size of the compared 10A IPM.

III. Conclusions

The new CIPOS™ Maxi is developed with smallest package compared with existing 1200V IPMs. It is offers the chance for costs saving and miniaturization with improved safety and simplified board design for the inverter system.

References

[1] R. Keggenhoff, Z.Liang, A. Arens, P. Kanschat, R. Rudolf. "Novel SOI Driver for Low Power Drive Applications", *Power Systems Design Europe, Nov. 2005*.

[2] Taehyun Kim, Minsub Lee, Junbae Lee "Protection Features of Intelligent Power Module against Transient State" *PCIM Europe, 2016*